Homework 6

Problem 1

给定下列 $f(n),g(n)$，判断 $f(n)=\mathrm{O}(g(n)),f(n)=\Omega(g(n))$ 是否成立
(1)$f(n)=2n^3+3n,g(n)=100n^2+2n+100$
(2)$f(n)=50n+\log n,g(n)=10n+\log\log n$
(3)$f(n)=50n\log n,g(n)=10n\log\log n$
(4)$f(n)=\log n,g(n)=\log^2 n$
(5)$f(n)=n!,g(n)=5^n$

Answer:

(1)$f(n)=\Omega(g(n))$
(2)$f(n)=\mathrm{O}(g(n)),f(n)=\Omega(g(n))$
(3)$f(n)=\Omega(g(n))$
(4)$f(n)=\mathrm{O}(g(n))$
(5)$f(n)=\Omega(g(n))$

Problem 2

举例增函数 $f(n),g(n)$，使得 $f(n)\neq\mathrm{O}(g(n)),f(n)\neq\Omega(g(n))$

Answer:

$$f(n)=n^{n+\sin n},g(n)=n^{n+\cos n}$$

Problem 3

按照增长速度排序以下函数

(1)$n\ln n$ (2)$(\ln\ln n)^{\ln n}$ (3)$(\ln n)^{\ln\ln n}$
(4)$n\cdot e^{\sqrt{\ln n}}$ (5)$(\ln n)^{\ln n}$ (6)$n\cdot 2^{\ln\ln n}$
(7)$n^{1+\frac{1}{\ln\ln n}}$ (8)$n^{1+\frac{1}{\ln n}}$ (9)n^2

Answer:

取对数

(1)$\ln n + \ln\ln n$
(2)$\ln n \cdot \ln\ln\ln n$
(3)$\ln\ln n \cdot \ln\ln n$
(4)$\ln n + \sqrt{\ln n}$
(5)$\ln n \cdot \ln\ln n$
(6)$\ln n \cdot \ln 2 \ln\ln n$
(7)$(1 + \frac{1}{\ln\ln n}) \ln n = \ln n + \frac{\ln n}{\ln\ln n}$
(8)$(1 + \frac{1}{\ln n}) \ln n = \ln n + 1$
(9)$2 \ln n$

借鉴快速排序的思路，以(1)为 `pivot`

(1)(2) $\ln n \cdot \ln\ln\ln n \succ 2\ln n \succ \ln n + \ln\ln n$ (2)≻(1)
(1)(3) $x^2 \prec e^x \therefore (\ln\ln n)^2 \prec e^{\ln\ln n} = \ln n \prec \ln n + \ln\ln n$ (3)≺(1)
(1)(4) $\sqrt{x} \succ \ln x \therefore \sqrt{\ln n} \succ \ln\ln n$ (4)≻(1)
(1)(5) $\ln n \cdot \ln\ln n \succ 2\ln n \succ \ln n + \ln\ln n$ (5)≻(1)
(1)(6) $\ln 2 \cdot \ln\ln n \prec \ln\ln n$ (6)≺(1)
(1)(7) $\ln n = e^{\ln\ln n} \succ (\ln\ln n)^2 \therefore \frac{\ln n}{\ln\ln n} \succ \ln\ln n$ (7)≻(1)
(1)(8) $1 \prec \ln\ln n$ (8)≺(1)
(1)(9) $\ln n \succ \ln\ln n$ (9)≻(1)

$\therefore$ (3)(6)(8) $\prec$ (1) $\prec$ (2)(4)(5)(7)(9)

(3)(6)(8)：以(3)为 `pivot`

(3)(6) $\ln n \succ \ln\ln n$ (6)≻(3)
(3)(8) $\ln n \succ (\ln\ln n)^2$ (8)≻(3)
(6)(8) $1 \prec \ln 2 \ln\ln n$ (8)≺(6)

$\therefore$ (3) $\prec$ (8) $\prec$ (6)

(2)(4)(5)(7)(9)：以(2)为 `pivot`

(2)(4) $\ln n \cdot \ln\ln\ln n \succ 2\ln n \succ \ln n + \sqrt{\ln n}$ (2)≻(4)
(2)(5) $\ln\ln\ln n \prec \ln\ln n$ (2)≺(5)
(2)(7) $\ln n \cdot \ln\ln\ln n \succ 2\ln n \succ \ln n + \frac{\ln n}{\ln\ln n}$ (2)≻(7)
(2)(9) $\ln n \cdot \ln\ln\ln n \succ 2\ln n$ (2)≻(9)

$\therefore$ (4)(7)(9) $\prec$ (2) $\prec$ (5)

(4)(7)(9)：以(4)为 `pivot`

(4)(7) $\sqrt{\ln n} \succ \ln\ln n \therefore \frac{\sqrt{\ln n}}{\ln\ln n} \succ 1 \therefore \frac{\ln n}{\ln\ln n} \succ \sqrt{\ln n}$ (7)≻(4)
(4)(9) $\ln n \succ \sqrt{\ln n}$ (9)≻(4)
(7)(9) $\ln n \succ \frac{\ln n}{\ln\ln n}$ (9)≻(7)

$\therefore$ (4) $\prec$ (7) $\prec$ (9)

最终排序(3)(8)(6)(1)(4)(7)(9)(2)(5)

Problem 4

若 $f_1(n) = \mathrm{O}(g_1(n)), f_2(n) = \mathrm{O}(g_2(n))$,
证明 $f_1(n) + f_2(n) = \mathrm{O}(g_1(n) + g_2(n)), f_1(n)f_2(n) = \mathrm{O}(g_1(n)g_2(n))$

Answer:

根据定义有

$$\begin{gathered}\forall n > N_1, f_1(n) \le c_1 g_1(n) \\ \forall n > N_2, f_2(n) \le c_2 g_2(n)\end{gathered}$$

所以

$$\begin{gathered}\forall n > \max\{N_1, N_2\}, f_1(n) + f_2(n) \le c_1 g_1(n) + c_2 g_2(n) \le \max\{c_1, c_2\}(g_1(n) + g_2(n)) \\ \forall n > \max\{n_1, N_2\}, c = c_1 c_2, f_1(n) f_2(n) \le c_1 c_2 g_1(n) g_2(n) = c g_1(n) g_2(n)\end{gathered}$$

即证 $f_1(n) + f_2(n) = \mathrm{O}(g_1(n) + g_2(n)), f_1(n)f_2(n) = \mathrm{O}(g_1(n)g_2(n))$

Problem 5

证明：$2\sqrt{n+1} - 2 < \sum_{i=1}^{n} \frac{1}{\sqrt{n}} < 2\sqrt{n} - 1$

Answer:

$$\begin{gathered}\sum_{i=1}^{n} \frac{1}{\sqrt{n}} > \int_1^{n+1} \frac{1}{\sqrt{x}} \mathrm{d}x = 2\sqrt{x}\Big|_1^{n+1} = 2\sqrt{n+1} - 2 \\ \sum_{i=1}^{n} \frac{1}{\sqrt{n}} = 1 + \sum_{i=2}^{n} \frac{1}{\sqrt{n}} < 1 + \int_1^{n} \frac{1}{\sqrt{x}} \mathrm{d}x = 1 + 2\sqrt{x}\Big|_1^{n} = 2\sqrt{n} - 1\end{gathered}$$